

Ornith-1.0 is a new family of open-source, self-improving models for agentic coding tasks, ranging from a 9B dense model for edge deployment to a 397B mixture-of-experts (MoE) flagship. Built on Gemma 4 and Qwen 3.5, Ornith-1.0 jointly learns both task solutions **and** the scaffolding (harnesses) that guide those solutions, enabling it to discover more effective search trajectories.

Key Highlights

- **Performance**
 - 397B version matches/exceeds Claude Opus 4.7 on Terminal-Bench 2.1 (77.5 vs 70.3) and SWE-Bench Verified (82.4 vs 80.8).
 - 35B variant outperforms similarly sized models and even beats larger Qwen 3.5-397B on Terminal-Bench.
 - 9B edge model delivers strong results (43.1 on Terminal-Bench, 69.4 on SWE-Bench) comparable to much larger models.
- **Self-Improving Training**
 - Each RL step first refines the scaffold, then generates a solution conditioned on it; rewards propagate to both stages, allowing scaffolds to evolve automatically.
- **Safety & Reward-Hacking Mitigations**
 1. Fixed outer trust boundary (environment, tool surface, test isolation).

2. Deterministic monitor flags forbidden actions, assigning zero reward.
 3. Frozen LLM judge acts as a veto on top of the verifier.
 - **Asynchronous RL**
 - Uses pipeline-RL with staleness weighting to handle long rollouts, down-weighting older tokens.
- https://deep-reinforce.com/ornith_1_0.html